1 Varieties

GHB - 538 (high rancid)



GHB - 732 (low rancid)



2 Variant calling

GHB - 538 (high rancid)

Initial total SNPs and INDELs: **17,842,802**

77.3%

Final total SNPs and INDELs: **13,799,108**

GHB - 732 (low rancid)

Initial total SNPs and INDELs: **18,686,050**

80.88%

Final total SNPs and INDELs: **15,112,744**

3 Variant breakdown

GHB - 538 (high rancid)

89.54% SNPs 10.46% INDELs

Final SNPs: **12,355,115**
Final INDELs: **1,443,993**

GHB - 732 (low rancid)

90.14% SNPs 9.86% INDELs

Final SNPs: **13,622,513**
Final INDELs: **1,490,231**

4 Genomic effect of Variants

GHB - 538 (high rancid)

35,772 classified gene id entry

- Gene having only SNP = **284**
- Gene having only INDEL = **21**
- Gene having SNP + INDEL = **35,466**

GHB - 732 (low rancid)

35,745 classified gene id entry

- Gene having only SNP = **280**
- Gene having only INDEL = **34**
- Gene having SNP + INDEL =**35,431**

5 Variant effect categories

Effect category	GHB - 538 (High rancid)	GHB - 732 (Low rancid)
Loss of function	13,336	10,181
Protein-altering	15,792	16,584
Non-coding / regulatory-context	4,300	5,551
Low impact coding	2,343	3,414
Intergenic-only entries	0	14
Compound gene ID	1	1
Total entries	35,772	35,745

7 Rancidity-associated genes

Summary	GHB - 538 (High rancid)		GHB - 732 (low rancid)	
	Category	Number of genes	Category	Number of genes
Total genes	1108		585	
	Protein-altering	497	Protein-altering	293
	LoF	419	LoF	149
	Non-coding / regulatory-context	98	Non-coding / regulatory-context	78
	Low impact coding	94	Low impact coding	65

6 Functional annotation percentage

Annotation category	GHB - 538 (High rancid)	GHB - 732 (Low rancid)
LoF	88.21%	74.14%
Protein-altering	92.17%	85.00%
Non-coding / regulatory-context	89.98%	86.78%
Low impact coding	94.24%	94.08%
Intergenic-only entries	-	100%
Compound gene ID	100%	100%
Overall annotation	90.56%	83.06%